

Building Technology

Basic Mathematics II for Building Technology

Professional Learner Notes

Geometry, mensuration, graphs, statistics and building measurement applications.

Section	Purpose
Plane geometry	Rectangles, triangles, trapezia and circles.
Volumes	Concrete, excavation and storage volumes.
Graphs	Straight-line graphs and building data.
Statistics	Averages and spread of production records.
Applied measurement	Drawings, schedules and quantity checks.

How to Study This Topic

The learning method follows a programmed engineering mathematics approach: understand the concept, study the formula, follow a worked method, then practise with a technical problem. The aim is not memorisation only; the learner must interpret the answer in the department context.

- Write the formula before substitution.
- Use correct units.
- Check whether the result is reasonable.
- State a technical interpretation.
- Repeat practice until the method is fluent.

Formula Summary

Rectangle area

$$A=lw$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

Volume

$$V=lwh$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

Circle area

$$A=\pi r^2$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

Cylinder volume

$$V=\pi r^2h$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

Mean

$$\bar{x}=\frac{\sum x}{n}$$

Meaning: identify each variable, check the units, substitute carefully, and state the final answer in the correct engineering context.

1. Plane geometry

Rectangles, triangles, trapezia and circles.

In building technology, this section is important because trainees use the calculation to make practical decisions in drawings, site measurements, setting out, material estimation, survey observations, quality checks, or revision work.

Key learning points

- Identify the physical quantity being calculated.
- Separate given values from required values.
- Select a formula that matches the problem.
- Use consistent units before calculation.
- Interpret the result as a technician, not only as a number.

Main formula for this section

$$A=lw$$

Worked method

Example: Floor slab. Calculate concrete volume for a slab 8 m by 5 m by 150 mm thick.

Step 1: Write the known quantities. Step 2: Write the formula. Step 3: Substitute the values.

Step 4: simplify carefully. Step 5: attach units. Step 6: write a one-sentence technical interpretation.

Engineering application

This method supports floor slabs. The result can be used to check whether a site measurement, drawing interpretation, material quantity, or survey computation is technically reasonable.

Common learner errors

- Using the wrong formula because the diagram/context was not analysed.
- Mixing units such as mm, cm and m.
- Rounding too early.
- Losing signs or angle direction.
- Submitting an answer without interpretation.

Practice set

1. Create and solve one technical calculation involving cylindrical tanks. Show formula, substitution, units and interpretation.
2. Create and solve one technical calculation involving block records. Show formula, substitution, units and interpretation.
3. Create and solve one technical calculation involving measurement schedules. Show formula, substitution, units and interpretation.
4. Create and solve one technical calculation involving floor slabs. Show formula, substitution, units and interpretation.
5. Create and solve one technical calculation involving excavation. Show formula, substitution, units and interpretation.

6. Create and solve one technical calculation involving cylindrical tanks. Show formula, substitution, units and interpretation.
7. Create and solve one technical calculation involving block records. Show formula, substitution, units and interpretation.
8. Create and solve one technical calculation involving measurement schedules. Show formula, substitution, units and interpretation.

2. Volumes

Concrete, excavation and storage volumes.

In building technology, this section is important because trainees use the calculation to make practical decisions in drawings, site measurements, setting out, material estimation, survey observations, quality checks, or revision work.

Key learning points

- Identify the physical quantity being calculated.
- Separate given values from required values.
- Select a formula that matches the problem.
- Use consistent units before calculation.
- Interpret the result as a technician, not only as a number.

Main formula for this section

$$V = lwh$$

Worked method

Example: Cylindrical tank. Find volume of a cylindrical tank with radius 0.7 m and height 1.8 m.

Step 1: Write the known quantities. Step 2: Write the formula. Step 3: Substitute the values.

Step 4: simplify carefully. Step 5: attach units. Step 6: write a one-sentence technical interpretation.

Engineering application

This method supports excavation. The result can be used to check whether a site measurement, drawing interpretation, material quantity, or survey computation is technically reasonable.

Common learner errors

- Using the wrong formula because the diagram/context was not analysed.
- Mixing units such as mm, cm and m.
- Rounding too early.
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Practice set

1. Create and solve one technical calculation involving block records. Show formula, substitution, units and interpretation.
2. Create and solve one technical calculation involving measurement schedules. Show formula, substitution, units and interpretation.
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7. Create and solve one technical calculation involving measurement schedules. Show formula, substitution, units and interpretation.
8. Create and solve one technical calculation involving floor slabs. Show formula, substitution, units and interpretation.

3. Graphs

Straight-line graphs and building data.

In building technology, this section is important because trainees use the calculation to make practical decisions in drawings, site measurements, setting out, material estimation, survey observations, quality checks, or revision work.

Key learning points

- Identify the physical quantity being calculated.
- Separate given values from required values.
- Select a formula that matches the problem.
- Use consistent units before calculation.
- Interpret the result as a technician, not only as a number.

Main formula for this section

$$A = \pi r^2$$

Worked method

Example: Production mean. Find average daily block production from five records.

Step 1: Write the known quantities. Step 2: Write the formula. Step 3: Substitute the values. Step 4: simplify carefully. Step 5: attach units. Step 6: write a one-sentence technical interpretation.

Engineering application

This method supports cylindrical tanks. The result can be used to check whether a site measurement, drawing interpretation, material quantity, or survey computation is technically reasonable.

Common learner errors

- Using the wrong formula because the diagram/context was not analysed.
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- Rounding too early.
- Losing signs or angle direction.
- Submitting an answer without interpretation.

Practice set

1. Create and solve one technical calculation involving measurement schedules. Show formula, substitution, units and interpretation.
2. Create and solve one technical calculation involving floor slabs. Show formula, substitution, units and interpretation.
3. Create and solve one technical calculation involving excavation. Show formula, substitution, units and interpretation.
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8. Create and solve one technical calculation involving excavation. Show formula, substitution, units and interpretation.

4. Statistics

Averages and spread of production records.

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- Identify the physical quantity being calculated.
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Main formula for this section

$$V = \pi r^2 h$$

Worked method

Example: Floor slab. Calculate concrete volume for a slab 8 m by 5 m by 150 mm thick.

Step 1: Write the known quantities. Step 2: Write the formula. Step 3: Substitute the values.

Step 4: simplify carefully. Step 5: attach units. Step 6: write a one-sentence technical interpretation.

Engineering application

This method supports block records. The result can be used to check whether a site measurement, drawing interpretation, material quantity, or survey computation is technically reasonable.

Common learner errors

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5. Applied measurement

Drawings, schedules and quantity checks.

In building technology, this section is important because trainees use the calculation to make practical decisions in drawings, site measurements, setting out, material estimation, survey observations, quality checks, or revision work.

Key learning points

- Identify the physical quantity being calculated.
- Separate given values from required values.
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- Interpret the result as a technician, not only as a number.

Main formula for this section

$$\bar{x} = \frac{\sum x}{n}$$

Worked method

Example: Cylindrical tank. Find volume of a cylindrical tank with radius 0.7 m and height 1.8 m.

Step 1: Write the known quantities. Step 2: Write the formula. Step 3: Substitute the values.

Step 4: simplify carefully. Step 5: attach units. Step 6: write a one-sentence technical interpretation.

Engineering application

This method supports measurement schedules. The result can be used to check whether a site measurement, drawing interpretation, material quantity, or survey computation is technically reasonable.

Common learner errors

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Practice set

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Mixed Revision Questions

1. A building technology problem involves excavation. Choose a suitable mathematical method, solve the problem, and explain the result.
2. A building technology problem involves cylindrical tanks. Choose a suitable mathematical method, solve the problem, and explain the result.
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30. A building technology problem involves floor slabs. Choose a suitable mathematical method, solve the problem, and explain the result.

References and Learning Sources

These learner materials are original notes prepared for Mr Mwirigi Mathematics Training Hub. The topic sequencing and level are informed by standard engineering mathematics references and the uploaded curriculum/OS materials. No textbook pages or solution-manual pages have been reproduced.

1. John Bird, Engineering Mathematics, Newnes / Elsevier.
2. K. A. Stroud and Dexter J. Booth, Engineering Mathematics, Palgrave Macmillan.
3. Uploaded KNEC solved mathematics materials for examination-style orientation.
4. Civil Engineering, Building Technology and Land Survey curriculum and occupational-standard materials used for local alignment.